together. Each one gains by presenting a smaller surface area to the elements than he would on his own. A fish who swims obliquely behind another fish may gain a hydrodynamic advantage from the turbulence produced by the fish in front. This could be partly why fish school. A related trick concerned with air turbulence is known to racing cyclists, and it may account for the V-formation of flying birds. There is probably competition to avoid the disadvantageous position at the head of the flock. Possibly the birds take turns as unwilling leader—a form of the delayed reciprocal altruism to be discussed at the end of the chapter.

Many of the suggested benefits of group living have been concerned with avoiding being eaten by predators. An elegant formulation of such a theory was given by W. D. Hamilton, in a paper called ‘Geometry for the Selfish Herd’. Lest this lead to misunderstanding, I must stress that by ‘selfish herd’ he meant ‘herd of selfish individuals’.

Once again we start with a simple ‘model’ which, though abstract, helps us to understand the real world. Suppose a species of animal is hunted by a predator that always tends to attack the nearest prey individual. From the predator’s point of view this is a reasonable strategy, since it tends to cut down energy expenditure. From the prey’s point of view it has an interesting consequence. It means that each prey individual will constantly try to avoid being the nearest to a predator. If the prey can detect the predator at a distance, it will simply run away. But if the predator is apt to turn up suddenly without warning, say it lurks concealed in long grass, then each prey individual can still take steps to minimize its chance of being the nearest to a predator. We can picture each prey individual as being surrounded by a ‘domain of danger’. This is defined as that area of ground in which any point is nearer to that individual than it is to any other individual. For instance, if the prey individuals march spaced out in a regular geometric formation, the domain of danger round each one (unless he is on the edge) might be roughly hexagonal in shape. If a predator happens to be lurking in the hexagonal domain of danger surrounding individual *A*, then individual *A* is likely to be eaten. Individuals on the edge of the herd are especially vulnerable, since their domain of danger is not a relatively small hexagon, but includes a wide area on the open side.

Now clearly a sensible individual will try to keep his domain of danger as small as possible. In particular, he will try to avoid being on the edge of the herd. If he finds himself on the edge he will take